

Transactions can be signed and aggregated in a form resistant to malicious actors but are not directly posted to the blockchain unless there is a discrepancy of some kind. This removes the constraints of a fixed block size and block rate, allowing for much higher throughput. Some layer-2 solutions are live today.

As Ethereum's transaction fees have risen to very high levels, layer-2 usage has remained stagnant. The space has been developing slowly, and many live solutions lack support for smart contracts or decentralized exchanges. An *optimistic rollup*, one idea under development, is a process in which transactions are aggregated off chain into a single digest that is periodically submitted to the chain over a certain interval. Only an aggregator who has a bond (stake) can combine and submit these summaries. Importantly, the state is assumed to be valid unless someone challenges it. If a challenge occurs, cryptography can prove if the aggregator posted a faulty state. The prover is then rewarded with a portion of the malicious aggregator's bond as an incentive (similar to a keeper mechanism). Optimistic rollups, though promising, have yet to deliver functional mainnets and require expensive fraud proofs as well as frequent rollup transaction posting, limiting their throughput and increasing their average transaction costs.

Many approaches aim to decrease the scalability risks facing DeFi today, but the field lacks a clear winner. As long as DeFi's growth is limited by blockchain scaling, applications will be limited in their potential impact.